

Foundation Programme
Universiti Teknologi Malaysia
Session 2019/2020 Semester 3

Test 2 (20%)

Time: 90 minutes

FSPM 0024 Calculus

Answer all SEVEN questions. List of formula is given on the next page.

1. If $\int_0^{\pi} f(x) dx = 3$ and $\int_{\pi}^{2\pi} f(x) dx = 6$, evaluate

(a) $\int_0^{2\pi} f(x) + x dx.$

(3 marks)

(b) $\int_{2\pi}^{\pi} 2f(x) - \sin x dx.$

(3 marks)

2. Find the values of A, B and C which satisfy

$$\frac{x^2 + 2x + 6}{(1-x)(x^2 + 2)} = \frac{A}{1-x} + \frac{Bx + C}{x^2 + 2}.$$

Hence, find $\int \frac{x^2 + 2x + 6}{(1-x)(x^2 + 2)} dx.$

(6 marks)

3. Evaluate $\int (x^3 + x) \sin 2x dx.$

(5 marks)

4. By using the given substitution, evaluate

$$\int e^x(e^x - 1)\sqrt{e^x + 1} dx, \quad u = e^x + 1.$$

(6 marks)

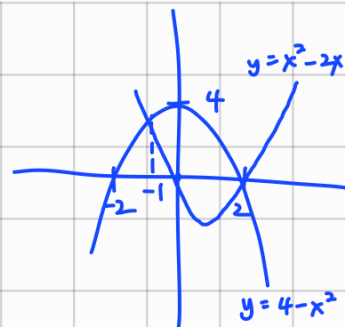
5. By using the substitution $x = 2 \sec \theta$, show that

$$\int_2^4 \frac{dx}{x^2 \sqrt{x^2 - 4}} = \frac{1}{4} \int_0^{\frac{\pi}{3}} \cos \theta d\theta.$$

Hence evaluate $\int_2^4 \frac{dx}{x^2 \sqrt{x^2 - 4}}.$

(7 marks)

$$\begin{aligned}
 (b) \quad 2 \int_0^2 \ln(x^{2x}) &= 4 \int_0^2 x \ln(x) dx = 4 \left(\frac{x^2 \ln x}{2} - \int \frac{1}{2} x dx \right) \\
 u &= \ln(x) \quad \frac{du}{dx} = \frac{1}{x} &= 4 \left(\frac{x^2 \ln x}{2} - \frac{1}{2} \int x dx \right) \\
 \frac{du}{dx} &= \frac{1}{x} \quad v = \frac{x^2}{2} &= 4 \left(\frac{x^2 \ln x}{2} - \frac{x^2}{4} \right) \\
 & &= 2x^2 \ln x - x^2 + C
 \end{aligned}$$



$$1) \int_0^\pi f(x) dx = 3, \int_\pi^{2\pi} f(x) dx = 6$$

$$\begin{aligned}
 (a) \quad \int_0^{2\pi} f(x) dx &= \int_\pi^{2\pi} f(x) dx + \int_0^\pi f(x) dx \quad \int_0^{2\pi} x dx = \left[\frac{x^2}{2} \right]_0^{2\pi} \\
 &= 6 + 3 &= 19.74 \\
 &= 9
 \end{aligned}$$

$$9 + 19.74 = 28.74$$

$$\begin{aligned}
 (b) \quad \int_{2\pi}^\pi 2f(x) - \sin x dx &= -2 \int_\pi^{2\pi} f(x) dx - \int_{2\pi}^\pi \sin x dx \\
 &= -2(6) + [\cos x]_{2\pi}^\pi \\
 &= -12 - 2 \\
 &= -14
 \end{aligned}$$

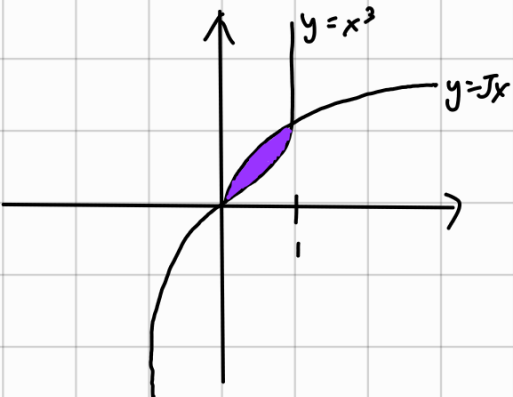
$$\begin{aligned}
 2) \quad A(x^2+2) + Bx + C(1-x) &= x^2 + 2x + 6 & 2B + C &= 4 \\
 (A+B)x^2 + (B+C)x + (A+C) &= x^2 + 2x + 6 & (1) \quad B-C &= 2 \\
 A+B &= 1 & 3B &= 6 \\
 B-C &= 2 & B &= 2 \\
 2A+C &= 6 & C &= 2 \\
 A &= 1+B & & \\
 A &= 3 & & \\
 2B+C &= 4 & & \\
 C &= 0 & &
 \end{aligned}$$

$$\begin{aligned}
 \int \frac{3}{1-x} + \frac{2x}{x^2+2} dx &= \int \frac{3}{1-x} dx + \int \frac{2x}{x^2+2} dx \quad u = x^2+2, \quad \frac{du}{dx} = 2x, \quad dx = \frac{du}{2x} \\
 &= 3 \ln(1-x) + \int \frac{1}{u} du \\
 &= 3 \ln(1-x) + \ln(x^2+2) + C
 \end{aligned}$$

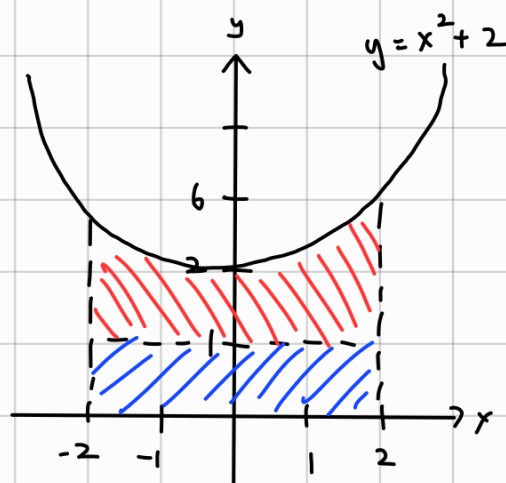
$$\begin{array}{lcl}
 3) \quad \int (x^3+x) \sin 2x dx & \begin{array}{l} \text{sign} \\ + \\ - \\ + \\ - \\ + \end{array} & \begin{array}{l} \text{differentiate } u \\ x^3+x \\ 3x^2+1 \\ 6x \\ 6 \\ 0 \end{array} \quad \begin{array}{l} \text{integrate } \frac{dv}{dx} \\ \sin 2x \\ -\frac{\cos 2x}{2} \\ -\frac{\sin 2x}{4} \\ \frac{\cos 2x}{8} \\ \frac{\sin 2x}{16} \end{array} \\
 u = x^3+x \quad \frac{du}{dx} = \sin 2x & & \int (x^3+x) \sin 2x dx \\
 & & = -\frac{(x^3+x)\cos 2x}{2} + \frac{(3x^2+1)\sin 2x}{4} + \\
 & & \frac{6x\cos 2x}{8} - \frac{6\sin 2x}{16} + C
 \end{array}$$

$$\begin{aligned}
 4) \int e^x(e^x-1)\sqrt{e^x+1} dx, \quad u &= e^x+1, \quad \frac{du}{dx} = e^x \\
 &= \int e^x(u)(\sqrt{u}) \cdot \frac{du}{e^x} \quad dx = \frac{du}{e^x} \\
 &= \int u^{\frac{3}{2}} du \\
 &= \frac{2}{5} u^{\frac{5}{2}} + C
 \end{aligned}$$

$$\begin{aligned}
 5) \int_2^4 \frac{1}{x^2 \sqrt{x^2-4}} dx &= \int_2^4 \frac{1}{4 \sec^2 \theta \sqrt{4(\sec^2 \theta - 1)}} \cdot 2 \sec \theta \tan \theta \cdot d\theta & \frac{1}{4} \int_0^{\frac{\pi}{3}} \cos \theta d\theta \\
 x &= 2 \sec \theta & = \frac{1}{4} [\sin \theta]_0^{\frac{\pi}{3}} \\
 \frac{dx}{d\theta} &= 2 \sec \theta \tan \theta & = \frac{1}{4} (\sin(\frac{\pi}{3}) - \sin 0) \\
 dx &= 2 \sec \theta \tan \theta d\theta & = \frac{1}{4} (\frac{\sqrt{3}}{2}) \\
 x=4, \theta &= \frac{\pi}{3} & = \frac{\sqrt{3}}{8} \\
 x=2, \theta &= 0
 \end{aligned}$$

$$\begin{aligned}
 6) \quad y &= \sqrt{x}, y = x^3, x=0, x=1 & \text{Area} &= \int_0^1 \sqrt{x} - x^3 dx \\
 & & &= \left[\frac{2}{3} x^{\frac{3}{2}} \right]_0^1 - \left[\frac{x^4}{4} \right]_0^1 \\
 & & &= \left(\frac{2}{3} - 0 \right) - \left(\frac{1}{4} - 0 \right) \\
 & & &= \frac{5}{12} \text{ unit}^2
 \end{aligned}$$


$$7) y = x^2 + 2, x = -2, x = 2, y = 1$$



$$\begin{aligned}
 \text{Volume} &= \pi \int_0^2 (x^2+2)^2 dx \\
 &= \pi \int_0^2 x^4 + 4x^2 + 4 dx \\
 &= \pi \left[\frac{x^5}{5} + \frac{4x^3}{3} + 4x \right]_0^2 \\
 &= \pi \left(\frac{32}{5} + \frac{32}{3} + 8 \right) \\
 &= 25.07 \pi \text{ unit}^3
 \end{aligned}$$

$$\begin{aligned}
 \pi r^2 h &= \pi (1^2)(4) \\
 &= 4\pi \text{ unit}^3
 \end{aligned}$$

$$\text{volume bounded} = 21.07 \pi \text{ unit}^3$$

6. Sketch and find the area of a region bounded by the curves $y = \sqrt{x}$, and $y = x^3$.

(5 marks)

7. Sketch the region that is bounded by the curve $y = x^2 + 2$, lines $x = -2$, $x = 2$, and $y = 1$. Find the volume of solid revolution formed by revolving the region 360° about the x -axis.

(5 marks)

LIST OF FORMULA

Trigonometric	
• $\cos^2 x + \sin^2 x = 1$ • $1 + \tan^2 x = \sec^2 x$ • $\cot^2 x + 1 = \operatorname{cosec}^2 x$ • $\sin 2x = 2 \sin x \cos x$ • $\cos 2x = \cos^2 x - \sin^2 x$ $= 2 \cos^2 x - 1$ $= 1 - 2 \sin^2 x$ • $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$	• $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ • $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ • $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ • $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$ • $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$ • $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$
Differentiations	Integrations
$\frac{d}{dx}[k] = 0, k \text{ constant}$	$\int k dx = kx + C, k \text{ constant}$
$\frac{d}{dx}[x^n] = nx^{n-1}$	$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$
$\frac{d}{dx}[e^x] = e^x$	$\int e^x dx = e^x + C$
$\frac{d}{dx}[\ln x] = \frac{1}{x}$	$\int \frac{dx}{x} = \ln x + C$
$\frac{d}{dx}[\cos x] = -\sin x$	$\int \sin x dx = -\cos x + C$
$\frac{d}{dx}[\sin x] = \cos x$	$\int \cos x dx = \sin x + C$
$\frac{d}{dx}[\tan x] = \sec^2 x$	$\int \sec^2 x dx = \tan x + C$
$\frac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x$	$\int \operatorname{cosec}^2 x dx = -\cot x + C$
$\frac{d}{dx}[\sec x] = \sec x \tan x$	$\int \sec x \tan x dx = \sec x + C$
$\frac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x$	$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$